EXPERIMENT – 3
FRAME-BASED KNOWLEDGE REPRESENTATION

1. Aim
To represent an object as a frame containing slots and their corresponding values using Python.

2. Software Required
Python 3.x

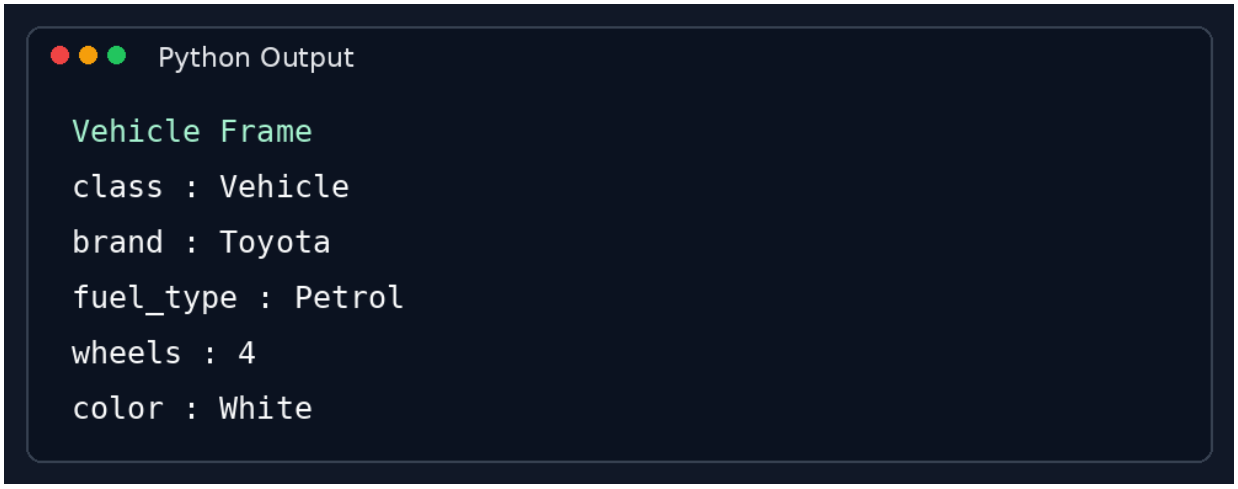
3. Procedure / Algorithm
1. Start the program.
2. Create a Python dictionary named vehicle to represent the frame.
3. Use dictionary keys as slots, such as class, brand, fuel_type, wheels, and color.
4. Assign an appropriate value to each slot.
5. Print the heading “Vehicle Frame”.
6. Use a for loop with items() to display each slot and its value.
7. Stop the program.

4. Python Program
```
vehicle = {
    "class": "Vehicle",
    "brand": "Toyota",
    "fuel_type": "Petrol",
    "wheels": 4,
    "color": "White"
}

print("Vehicle Frame")
for slot, value in vehicle.items():
    print(slot, ":", value)
```

5. Sample Input
The frame values are predefined in the program. No keyboard input is required.

6. Output Screenshot
The following screenshot-style image shows the expected console output:



7. Expected Output

Vehicle Frame
class : Vehicle
brand : Toyota
fuel_type : Petrol
wheels : 4
color : White

8. Result

The vehicle was successfully represented using a frame in the form of a Python dictionary. Each slot stores a corresponding value, and the program displays all slots and values correctly.